

The digit-weight law for Weyl operators in prime-power dimension: sparsity versus nullity up to sparsity $2p - 1$, exact root-of-unity counts, and binomial and trinomial laws in every dimension

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Abstract

Let $d = p^k$ and let $W_{(a,b)} = X^a Z^b$ be the cyclic Weyl operators on $\mathbb{C}^d$. Write $w_p(r)$ for the base- p digit weight $\prod_t (r_t + 1)$ and $\Omega_k(m) = \max\{r < p^k : w_p(r) \leq m\}$. We prove: (Z) a polynomial with at most m terms and degree below d annihilates at most $\Omega_k(m)$ of the d -th roots of unity, and this is sharp; (B) every nonzero combination C of $\ell \leq 2p - 1$ Weyl operators satisfies $\dim \ker C \leq \Omega_k(\ell)$, i.e. $\text{rank } C \geq p^k - (\ell - 1)p^{k-1}$ for $\ell \leq p$ (sharp) and $\text{rank } C \geq p^{k-1}$ for $p \leq \ell \leq 2p - 1$. The commutative ingredient is the Massey–Costello–Justesen weight-retaining property; the operator ingredient is a Radon-type lemma showing that first-order obstructions have weight at least $2p$. For composite d the maximal nullity is $d/p_{\min}$ at sparsity two and, conditionally on a published trinomial rank rule, $\max(2d/r_3, d/2 [d \text{ even}])$ at sparsity three. The conjecture $\dim \ker C \leq \Omega_k(s_W(C))$ for all sparsities is proved for $d \in \{4, 8, 9\}$ and for torsion cosets; its first open case is $d = 16, \ell = 6$.

1. Introduction

1.1 The results

Fix $d = p^k$, $\omega = e^{2\pi i/d}$, the cyclic Weyl basis $W_{(a,b)} = X^a Z^b$ of $M_d(\mathbb{C})$ indexed by $G = (\mathbb{Z}/d)^2$, and for $C = \sum_g c_g W_g$ the Weyl sparsity $s_W(C) = \#\{g : c_g \neq 0\}$. The published prime law [3W] states that in prime dimension $\dim \ker C \leq s_W(C) - 1$. Its Open question 3 asked what replaces this in prime-power dimension, where the coranks $p^j, 2p^j$ of [0WE] Theorem 3 already show that $\ell - 1$ is wrong, and suggested the candidate $(\ell - 1)p^{k-1}$. This paper answers the question in the range $\ell \leq 2p - 1$ and identifies the function that governs the problem. With $w_p(r) = \prod_t (r_t + 1)$ for $r = \sum_t r_t p^t$ and

$$\Omega_k(m) = \max\{r < p^k : w_p(r) \leq m\},$$

the main results are:

- **Theorem Z (roots of unity).** The maximal number of p^k -th roots of unity annihilated by a nonzero polynomial of degree $< p^k$ with at most m nonzero coefficients is exactly $\Omega_k(m)$. The extremal polynomials are explicit products, certified exactly in $\mathbb{Z}[\omega_d]$ for $d \in \{4, 8, 9, 16, 25, 27, 32, 49, 81\}$.

- **Theorem B (operators).** If $d = p^k$ and $1 \leq s_W(C) = \ell \leq 2p - 1$, then $\dim \ker C \leq \Omega_k(\ell)$. Explicitly, $\dim \ker C \leq (\ell - 1)p^{k-1}$ for $\ell \leq p$, sharp at every such ℓ , and $\dim \ker C \leq (p - 1)p^{k-1}$ for $p \leq \ell \leq 2p - 1$ ($k \geq 2$). The candidate $(\ell - 1)p^{k-1}$ of [3W] is therefore right exactly up to $\ell = p$; beyond it the true envelope is Ω_k , which grows much more slowly (for example $\Omega_k(p^a) = p^k - p^{k-a}$).
- **Theorems T2 and T3 (composite d).** For every $d \geq 2$ the maximal nullity at sparsity two is $d/p_{\min}(d)$; for every $d \geq 3$ the maximal nullity at sparsity three is $\max(2d/r_3(d), \frac{d}{2}[d \text{ even}])$, where $r_3(d)$ is the smallest divisor of d that is at least 3; the second branch is attained by an anticommuting pair when $d \equiv 2 \pmod{4}$. Theorem T3 is conditional on the trinomial rank rule [0WE] Theorem 2, used as published. Both laws refute every additive law in the largest prime or prime-power factor of d .

The commutative ingredient (Theorem A, Section 3) is a characteristic- p statement: a polynomial with m nonzero terms vanishes at $x = 1$ to order r only if $w_p(r) \leq m$. This is a special case of the weight-retaining property of Massey, Costello and Justesen [MCJ73], and we say so plainly; a short self-contained proof is included because the form needed here (arbitrary fields of characteristic p) is the one used in the valuation argument. The operator ingredient is new relative to [3W]: the pigeonhole over $p + 1$ directions of [3W] is replaced by Lemma N, which shows that a residue vector with vanishing fibre sums in all $p^{k-1}(p + 1)$ primitive directions has at least $2p$ nonzero entries, via a Radon-type lemma on $\mathbb{F}_p^2$ (Lemma R).

Beyond $2p - 1$ the first-order method genuinely stops (two parallel lines of a p -torsion coset form an obstruction of weight $2p$), and we state Conjecture C: $\dim \ker C \leq \Omega_k(s_W(C))$ for all C . It is a theorem for $d \in \{4, 8, 9\}$, for every support inside a coset of the p -torsion, for every support inside a coset of the 4-torsion when $d \in \{16, 32\}$, and wherever the field-free bound $\text{rank } C \cdot s_W(C) \geq d$ suffices; the first open case is $d = 16, \ell \in \{6, 7\}$ (Section 6).

1.2 Antecedents

The prime case is [3W] (Sections 2–5 there: the setting, the valuation reduction modulo $(1 - \omega)$, the explicit Clifford conjugations $F'Q_s$, the pigeonhole, and Frenkel’s lemma), which in turn rests on Tao’s uncertainty principle [Ta05] and Frenkel’s proof of Chebotarev’s theorem [Fr04]. The three-term record [0WE] proves the trinomial rank rule (Theorem 2 there, with its Lemma 1 on central sectors), the classification of coranks for $s_W \leq 3$ in prime-power dimension (Theorem 3), and exhibits composite counterexamples to any $\ell - 1$ law: $2^{1/4}I + X^3 + Z^3$ in $d = 12$ (nullity 3, Proposition 4), $I + X^3 + iX^{-3}Z^{18}$ in $d = 36$ (nullity 6, eq. (6a)), and the binomial $X(I + Z^6)$ in $d = 12$ (nullity 6, eq. (6c)). The origin of the Weyl rank problem is quantum list decoding: [3H0] determines exact list thresholds for Weyl channels with consecutive and arithmetic probes, [0WE] eq. (8) reduces list decoding to a rank problem, and [3W] Corollary 2 closes the prime case. The list-decoding consequences of the present rank bounds are not pursued here.

On the commutative side, Meshulam’s uncertainty inequality for finite abelian groups [Me06] is the general tool (used in Section 6 for torsion cosets); Delvaux and Van Barel [DVB08] describe the maximal rank-deficient submatrices of Fourier matrices of prime-power order through a hierarchical block structure, which is the closest antecedent to Theorem Z, although the zero-count function Ω_k is not stated there; Lam and Leung [LL00] classify the lengths of vanishing sums of roots of unity, the phenomenon behind the “smallest divisor” laws of Section 5; Bíró and Lev [BL18] give the sharp uncertainty inequality on $\mathbb{F}_p^2$, relevant to the p -torsion cosets of Section 6; Loukaki [Lo25] treats principal Fourier minors of composite order pq . Lucas’ theorem [Lu78] is what makes w_p the right weight.

1.3 Calibration

Theorem A is classical ([MCJ73]); the searches recorded in the working report found no statement of Theorem Z (the exact function Ω_k for root-of-unity counts) or of Theorem B (the operator law), but a bounded negative search does not certify priority, and the hierarchical structure in [DVB08] may contain Theorem Z implicitly. Theorems T2 and T3 are short consequences of trace orthogonality and of [0WE] Theorem 2 respectively. Every numerical and exact claim was recomputed by an independent adversarial reviewer with separately written code (Section 7); three expository gaps found by that review are closed in Section 4, and one labelling error of the working report (exact versus finite-field tables) is corrected in Section 5.

2. Notation and setting

Weyl operators. For $d \geq 2$ let $\omega = \omega_d = e^{2\pi i/d}$, $X|j\rangle = |j+1 \bmod d\rangle$, $Z|j\rangle = \omega^j|j\rangle$, so $ZX = \omega XZ$, and $W_{(a,b)} = X^a Z^b$ for $(a,b) \in G = (\mathbb{Z}/d)^2$. Then $W_g W_h = \omega^{\langle g,h \rangle} W_{g+h}$ with $\langle (a,b), (a',b') \rangle = ba'$, $\text{Tr } W_g = 0$ for $g \neq 0$, and the W_g form a trace-orthogonal basis of $M_d(\mathbb{C})$. For $C = \sum_g c_g W_g$ the support is $S(C) = \{g : c_g \neq 0\}$ and $s_W(C) = |S(C)|$. Left multiplication by W_h^{-1} translates the support by $-h$, multiplies coefficients by roots of unity and preserves the kernel; conjugation by an invertible matrix U with $UW_g U^{-1} \in \mathbb{C} W_{Mg}$ ($M \in SL_2(\mathbb{Z}/d)$) maps the support to MS and preserves rank. Two families of such U are made explicit in Section 4.3.

Digit weight. For $d = p^k$ and $0 \leq r < p^k$ write $r = \sum_{t=0}^{k-1} r_t p^t$ with $0 \leq r_t < p$ and put $w_p(r) = \prod_t (r_t + 1)$. By Lucas' theorem, $\binom{r}{i} \not\equiv 0 \pmod{p}$ exactly when every digit of i is at most the corresponding digit of r , so $w_p(r)$ is the number of nonzero terms of $(x-1)^r$ in characteristic p . For $m \geq 1$ let $\Omega_k(m) = \max\{r < p^k : w_p(r) \leq m\}$; it is nondecreasing in m , $\Omega_k(1) = 0$, $\Omega_k(p^k) = p^k - 1$, and

$$\begin{aligned} \Omega_k(m) &= (m-1)p^{k-1} & (1 \leq m \leq p), \\ \Omega_k(m) &= (p-1)p^{k-1} + ([m/p] - 1)p^{k-2} & (p \leq m < p^2, k \geq 2), \\ \Omega_k(p^a) &= p^k - p^{k-a} & (0 \leq a \leq k). \end{aligned}$$

(For $m \leq p$: a number above $(m-1)p^{k-1}$ has top digit $\geq m$, or top digit $m-1$ and a further nonzero digit, hence weight $> m$. The other two formulas are checked the same way; all three were also verified by computer for every $p^k \leq 2401$.) For example, for $d = 16$ the values $\Omega_4(1..16)$ are 0, 8, 8, 12, 12, 12, 12, 14, ..., 14, 15; for $d = 27$, $\Omega_3(m)$ is 0, 9, 18 for $m = 1, 2, 3$, then 18 ($m = 4, 5$), 21 ($6 \leq m \leq 8$), 24 ($9 \leq m \leq 17$), 25 ($18 \leq m \leq 26$), 26.

Torsion and directions. For $d = p^k$ let $T_j = p^{k-j}G$ be the p^j -torsion, so $T_1 \cong \mathbb{F}_p^2$. A *primitive direction* is a surjective linear form $\lambda : G \rightarrow \mathbb{Z}/d$ up to units; there are $|\mathbb{P}^1(\mathbb{Z}/p^k)| = p^{k-1}(p+1)$ of them, and the fibres of λ are the cosets of $\ker \lambda$, a cyclic subgroup of order p^k ; we call these cosets *primitive lines*. Every cyclic subgroup of order p^k is the kernel of exactly one primitive direction. A support contained in a primitive line is called *collinear*.

Valuation rings. Let K be a subfield of $\mathbb{C}$ containing ω_d . A valuation ring $\mathcal{O} \subset K$ with maximal ideal $\mathfrak{m}$, residue field $\mathbb{k} = \mathcal{O}/\mathfrak{m}$ and valuation v is *admissible* if $1 - \omega_d \in \mathfrak{m}$. Since $1 - \omega_{p^k}$ generates the unique prime of $\mathbb{Z}[\omega_{p^k}]$ above p and p is a unit multiple of $(1 - \omega_{p^k})^{\varphi(p^k)}$, admissibility forces $p \in \mathfrak{m}$, so $\text{char } \mathbb{k} = p$ and every d -th root of unity reduces to 1. Admissible valuation rings exist on every finitely generated $K \supseteq \mathbb{Q}(\omega_d)$ by Chevalley's extension theorem ([St, Tag 00IA]; this is the route of [3W], Step 1). Two rank facts are used: for $A \in M_d(\mathcal{O})$, $\text{rank}_K A \geq \text{rank}_{\mathbb{k}} \bar{A}$; for $A \in M_d(K)$, $\text{rank}_K A = \text{rank}_{\mathbb{C}} A$.

Root-of-unity counts. For any d and $m \geq 1$ put $Z(d, m) = \max \#\{\zeta \in \mu_d : f(\zeta) = 0\}$ over polynomials f with exactly m nonzero coefficients and exponents distinct modulo d , and $Z_{\leq}(d, m) = \max_{m' \leq m} Z(d, m')$. Since $\{f(\zeta)\}_{\zeta \in \mu_d}$ is the spectrum of $f(Z)$, $Z(d, m)$ is the maximal nullity of a polynomial in Z with exactly m terms, i.e. of a collinear Weyl combination on the Z -axis with exact sparsity m .

3. Statements

Theorem A (order and digit weight in characteristic p ; [MCJ73]). Let $\mathbb{k}$ be any field of characteristic p and $f \in \mathbb{k}[x]$ nonzero with exactly m nonzero monomials. Then $w_p(\text{ord}_{x=1} f) \leq m$. If moreover $\deg f < p^k$, then $\text{ord}_1 f \leq \Omega_k(m)$, and $\max\{\text{ord}_1 f : \deg f < p^k, \# \text{terms} \leq m\} = \Omega_k(m)$, attained by $(x-1)^{\Omega_k(m)}$.

Theorem A is the weight-retaining property of Massey, Costello and Justesen ([MCJ73], Theorem 1.1 for $\text{GF}(2^r)$ and Section III for $\text{GF}(q)$, with their Lemma 1 giving $W[(x-c)^i] = \prod_j (i_j + 1)$): writing $f = \sum_{i \geq r} b_i (x-1)^i$ with $b_r \neq 0$ and $r = \text{ord}_1 f$, their theorem states that the Hamming weight of f is at least that of $(x-1)^r$. They state it over finite fields; the proof in Section 4.1 is written for an arbitrary field of characteristic p , which is the form the valuation argument needs. The degree restriction is needed only for the reformulation through Ω_k (note $x^{p^k} - 1 = (x-1)^{p^k}$ has two terms).

Theorem Z (roots of unity in prime-power order). For $d = p^k$ and $m \geq 1$, $Z_{\leq}(d, m) = \Omega_k(m)$. Equivalently, a polynomial in Z with at most m terms has nullity at most $\Omega_k(m)$, and for every $r < p^k$ the polynomial

$$f_r(z) = \prod_{j=1}^k \prod_{i=1}^{r_{j-1}} (z^{p^{j-1}} - \omega_p^i), \quad \omega_p = \omega^{p^{k-1}} = e^{2\pi i/p}, \quad r = \sum_{j=1}^k r_{j-1} p^{j-1},$$

has exactly $w_p(r)$ nonzero coefficients, degree r , and exactly r zeros in μ_d . In particular $Z(d, m) = \Omega_k(m)$ whenever $w_p(\Omega_k(m)) = m$ (for instance $m \leq p$), and $Z(d, m) \leq \Omega_k(m)$ always.

Theorem B' (valuation form, all sparsities). Let $d = p^k$, let $K \subseteq \mathbb{C}$ contain ω_d and, if $p = 2$, also ω_{2d} ; let $\mathcal{O} \subset K$ be admissible. Let $C = \sum_{g \in S} c_g W_g$ with all $c_g \in K^\times$, scaled so that $\min_g v(c_g) = 0$, and let $\bar{c}_g \in \mathbb{k}$ be the residues. For a primitive direction λ put $\gamma_t^\lambda = \sum_{\lambda(g)=t} \bar{c}_g$ ($t \in \mathbb{Z}/d$) and $m_\lambda = \#\{t : \gamma_t^\lambda \neq 0\}$. Then for every λ with $m_\lambda \geq 1$,

$$\dim \ker C \leq \Omega_k(m_\lambda) \leq \Omega_k(|S|).$$

Lemma R (Radon kernel on $\mathbb{F}_p^2$). Let $\mathbb{k}$ have characteristic p and let $f : \mathbb{F}_p^2 \rightarrow \mathbb{k}$ be nonzero with zero sum on every affine line. Then $|\text{supp } f| \geq 2p$. The bound is attained by $1_{L_1} - 1_{L_2}$ for two parallel lines.

Lemma N (first-order obstructions have weight at least $2p$). Let $d = p^k$ and let $\bar{c} : G \rightarrow \mathbb{k}$ be nonzero with zero sum on every primitive line. Then $|\text{supp } \bar{c}| \geq 2p$; two parallel lines inside a coset of T_1 attain the bound.

Theorem B (operator law in prime-power dimension, $\ell \leq 2p - 1$). Let $d = p^k$ and $C \neq 0$ with $s_W(C) = \ell \leq 2p - 1$. Then $\dim \ker C \leq \Omega_k(\ell)$; explicitly

$$\text{rank } C \geq p^k - (\ell - 1)p^{k-1} \quad (\ell \leq p), \quad \text{rank } C \geq p^{k-1} \quad (p \leq \ell \leq 2p - 1, k \geq 2).$$

For $\ell \leq p$ the bound is attained by $C_\ell = \prod_{i=0}^{\ell-2} (Z^{p^{k-1}} - \omega_p^i I)$, which has $s_W = \ell$ and nullity $(\ell - 1)p^{k-1}$. For $k = 1$ this is [3W] Theorem 1.

Corollary (coranks at sparsity ≤ 3). $\{r : w_p(r) \leq 3\} = \{0\} \cup \{p^j, 2p^j : j < k\}$ for odd p and $\{0\} \cup \{2^j : j < k\}$ for $p = 2$, so Theorem B' bounds the nullity of every C with $s_W(C) \leq 3$ by the largest element of this set, $2p^{k-1}$ resp. 2^{k-1} , in agreement with [0WE] Theorem 3, whose corank set $\mathcal{N}_p$ is exactly this set without 0.

Theorem T2 (binomials, every $d \geq 2$). $\max\{\dim \ker C : s_W(C) = 2\} = d/p_{\min}(d)$, where $p_{\min}(d)$ is the smallest prime factor of d .

Theorem T3 (trinomials, every $d \geq 3$; conditional on [0WE] Theorem 2). Let $r_3(d)$ be the smallest divisor of d that is ≥ 3 . Then

$$v_3(d) := \max\{\dim \ker C : s_W(C) = 3\} = \max\left(\frac{2d}{r_3(d)}, \frac{d}{2} [d \text{ even}]\right),$$

$$\max\{\dim \ker C : s_W(C) \leq 3\} = \max\left(\frac{d}{p_{\min}(d)}, \frac{2d}{r_3(d)}\right).$$

The second branch of v_3 is strictly larger than the first exactly when $d \equiv 2 \pmod{4}$ and $3 \nmid d$; it is attained by the anticommuting pair $Z^{d/2}, X^{d/2}$.

Conjecture C (prime powers, all sparsities). For $d = p^k$ and every nonzero C , $\dim \ker C \leq \Omega_k(s_W(C))$.

Conjecture U (every d). For every d and every nonzero C , $\dim \ker C \leq Z_{\leq}(d, s_W(C))$: the monotone collinear maximum bounds all operators. (The exact-sparsity version $\dim \ker C \leq Z(d, s_W(C))$ is false: $d = 10, \ell = 3$, Section 5.)

4. Proofs

4.1 Theorem A

Frenkel's lemma ([Fr04]; [3W] Step 6): a nonzero $q = \sum_{i=1}^n a_i x^{t_i} \in \mathbb{k}[x]$ with distinct exponents $t_i \in \{0, \dots, p-1\}$ has $\text{ord}_1 q \leq n-1$. Indeed, with $\theta = x d/dx$, θ maps $(x-1)^a \mathbb{k}[x]$ into $(x-1)^{a-1} \mathbb{k}[x]$, so $\text{ord}_1 q \geq n$ would give $(\theta^s q)(1) = \sum_i a_i t_i^s = 0$ for $0 \leq s < n$, a Vandermonde system in the t_i , which are distinct in $\mathbb{F}_p \subseteq \mathbb{k}$; hence all $a_i = 0$.

We prove $w_p(\text{ord}_1 f) \leq m$ by induction on k , where $\deg f < p^k$ (every polynomial has such a k ; the degree restriction is only used to run the induction). For $k = 0$, f is a nonzero constant, $\text{ord}_1 f = 0$ and $w_p(0) = 1 \leq m$. For $k \geq 1$ write $f = \sum_{j=0}^{p-1} x^j g_j(x^p)$ with $\deg g_j < p^{k-1}$; the exponents $pj' + j$ are distinct, so $m = \sum_j \# \text{terms}(g_j)$. Put $x = 1 + y$. Since $(1+y)^p = 1 + y^p$,

$$f(1+y) = \sum_j (1+y)^j g_j(1+y^p) = \sum_{n \geq 0} y^{pn} P_n(y), \quad P_n(y) = \sum_j h_{j,n} (1+y)^j,$$

where $h_{j,n}$ is the coefficient of Y^n in $g_j(1+Y)$ (the n -th Hasse derivative of g_j at 1) and $\deg P_n \leq p-1$. The block $y^{pn} P_n$ occupies degrees $[pn, pn+p-1]$, disjoint for different n . Hence, if $r = \text{ord}_1 f = pr' + r_0$ with $0 \leq r_0 < p$, then $P_n = 0$ for $n < r'$ and $\text{ord}_y P_{r'} = r_0$. Because $\{(1+y)^j\}_{0 \leq j < p}$ is a basis of the polynomials of degree $< p$, $P_n = 0$ if and only if $h_{j,n} = 0$ for all j . Therefore $\text{ord}_1 g_j \geq r'$ for every j , with equality exactly on $J_0 = \{j : h_{j,r'} \neq 0\} \neq \emptyset$. The polynomial $q(x) = P_{r'}(x-1) = \sum_{j \in J_0} h_{j,r'} x^j$ is

nonzero, has $|J_0|$ terms with distinct exponents in $\{0, \dots, p-1\}$, and $\text{ord}_1 q = r_0$; Frenkel's lemma gives $|J_0| \geq r_0 + 1$. For $j \in J_0$ the polynomial g_j is nonzero, has degree $< p^{k-1}$ and order r' , so by induction $\#\text{terms}(g_j) \geq w_p(r')$. Altogether

$$m \geq \sum_{j \in J_0} \#\text{terms}(g_j) \geq |J_0| w_p(r') \geq (r_0 + 1) w_p(r') = w_p(r).$$

If $\deg f < p^k$ then $r < p^k$, so $w_p(r) \leq m$ means $r \leq \Omega_k(m)$; conversely $(x-1)^{\Omega_k(m)}$ has $w_p(\Omega_k(m)) \leq m$ terms (Lucas) and order $\Omega_k(m)$. $\square$

Remark. The function $m \mapsto \max\{\text{ord}_1 f : \text{exactly } m \text{ terms}\}$ is not monotone (for $d = 16$ it takes the value 8 at $m = 2$ and 4 at $m = 3$); the clean statement is the “at most m ” one.

4.2 Theorem Z

Upper bound. Let $f \in \mathbb{C}[x]$ be nonzero, $\deg f < d$, with at most m terms. Let $K = \mathbb{Q}(\omega)$ (coefficients of f), finitely generated over $\mathbb{Q}(\omega)$, and let $\mathcal{O}$ be an admissible valuation ring of K ; divide f by a coefficient of minimal valuation, so $f \in \mathcal{O}[x]$ and the reduction $\bar{f} \in \mathbb{k}[x]$ is nonzero with at most m terms and degree $< p^k$. The number of zeros of f in μ_d is the nullity of the diagonal matrix $f(Z)$; conjugating by the unnormalised Fourier matrix $F' = (\omega^{jl})$, for which $F'^{-1}ZF' = X$ (Section 4.3), it is also the nullity of the circulant $f(X) \in M_d(\mathcal{O})$. (Reducing $f(Z)$ itself would only give $\bar{f}(1)I$; the circulant form is what makes the reduction informative.) Now $\text{rank}_{\mathbb{C}} f(X) = \text{rank}_K f(X) \geq \text{rank}_{\mathbb{k}} \bar{f}(\bar{X})$, where $\bar{X}$ is the cyclic shift over $\mathbb{k}$. As a $\mathbb{k}[x]$ -module, $\mathbb{k}^d \cong \mathbb{k}[x]/(x^{p^k} - 1) = \mathbb{k}[x]/(x-1)^{p^k}$, and $\bar{f}(\bar{X})$ is multiplication by $\bar{f} = (x-1)^r u$ with $u(1) \neq 0$, $r = \text{ord}_1 \bar{f} \leq \deg \bar{f} < p^k$; u is a unit modulo $(x-1)^{p^k}$ and the kernel of multiplication by $(x-1)^r$ has dimension r . Hence $\#\{\zeta \in \mu_d : f(\zeta) = 0\} \leq r \leq \Omega_k(m)$ by Theorem A.

Lower bound. Fix $r < p^k$ with digits r_{j-1} and let f_r be as stated. For $n \leq p-1$ the q -binomial theorem with $q = \omega_p$ gives $\prod_{i=1}^n (u - \omega_p^i) = \sum_{t=0}^n (-1)^t \omega_p^{t(t+1)/2} \binom{n}{t}_{\omega_p} u^{n-t}$, and every Gaussian binomial $\binom{n}{t}_{\omega_p} = \prod_{i=1}^t (1 - \omega_p^{n-t+i}) / (1 - \omega_p^i)$ is nonzero because ω_p is a primitive p -th root of unity and all exponents are in $[1, p-1]$. So the j -th inner product has exactly $r_{j-1} + 1$ nonzero coefficients, at the exponents ip^{j-1} , $0 \leq i \leq r_{j-1}$. The exponents of the full product are the numbers $\sum_j i_j p^{j-1}$ with $0 \leq i_j \leq r_{j-1}$, which are pairwise distinct base- p expansions below p^k ; no cancellation occurs and f_r has exactly $\prod_j (r_{j-1} + 1) = w_p(r)$ terms and degree r . The factor $z^{p^{j-1}} - \omega_p^i$ ($1 \leq i \leq p-1$) vanishes at the $z \in \mu_d$ with $z^{p^{j-1}} = \omega_p^i$: the map $z \mapsto z^{p^{j-1}}$ from μ_{p^k} onto $\mu_{p^{k-j+1}}$ has fibres of size p^{j-1} and $\omega_p^i \in \mu_p \subseteq \mu_{p^{k-j+1}}$, so there are exactly p^{j-1} such z , all of exact order p^j . These sets are disjoint over (j, i) , so f_r has at least $\sum_j r_{j-1} p^{j-1} = r$ zeros in μ_d , and exactly r since $\deg f_r = r$. Taking $r = \Omega_k(m)$ gives $Z_{\leq}(d, m) \geq \Omega_k(m)$. $\square$

Note that modulo $1 - \omega$ each factor becomes $(z-1)^{p^{j-1}}$, so $\bar{f}_r = (z-1)^r$: the construction lifts the extremal polynomial of Theorem A.

4.3 The Clifford family and the $p = 2$ phases

Let $F' = (\omega^{jl})_{j,l}$, so $F'^{-1} = d^{-1}F'^*$, $F'XF'^{-1} = Z$, $F'ZF'^{-1} = X^{-1}$, hence $F'^{-1}XF' = Z^{-1}$ and $F'^{-1}ZF' = X$ (valid for every d). For odd d let 2^{-1} be the inverse of 2 modulo d and $Q_s = \text{diag}(\omega^{s \cdot 2^{-1} j^2})_j$ for $s \in \mathbb{Z}/d$; for $d = 2^k$ let $\omega_{2d} = e^{2\pi i/2d}$ and $Q_s = \text{diag}(\omega_{2d}^{s j^2})_j$ (well defined on $j \in \mathbb{Z}/d$ because $(j+d)^2 \equiv j^2 \pmod{2d}$). In both cases $Q_s Z Q_s^{-1} = Z$ and $Q_s X Q_s^{-1} = \phi_s X Z^s$ with $\phi_s = \omega^{s \cdot 2^{-1}}$ resp.

ω_{2d}^s , and using $(XZ^s)^a = \omega^{sa(a-1)/2} X^a Z^{sa}$ one gets $Q_s W_{(a,b)} Q_s^{-1} = \varepsilon X^a Z^{sa+b}$ with $\varepsilon = \omega^{s \cdot 2^{-1} a^2}$ resp. $\omega_{2d}^{sa^2}$. Put $U_s = F' Q_s$ and $V_s = F'^{-1} Q_s F'$. Then, exactly in $M_d(\mathbb{Q}(\omega))$ resp. $M_d(\mathbb{Q}(\omega_{2d}))$,

$$U_s W_{(a,b)} U_s^{-1} = \varepsilon \omega^{-a(sa+b)} W_{(-(sa+b), a)}, \quad V_s W_{(a,b)} V_s^{-1} = \varepsilon' W_{(a-sb, b)},$$

with $\varepsilon' = \omega^{-s \cdot 2^{-1} b^2}$ resp. $\omega_{2d}^{-sb^2}$ (compute $F' W_{(a,b)} F'^{-1} = \omega^{-ab} W_{(-b, a)}$, apply Q_s , and conjugate back). The first coordinates of the label maps are the linear forms $\lambda_s(a, b) = -(sa + b)$ for U_s ($s \in \mathbb{Z}/d$), $\lambda_\infty(a, b) = a$ for $U_\infty = I$, and $\lambda'_s(a, b) = a - sb$ for V_s . The forms $\lambda_s, \lambda_\infty$ give the $p^k + 1$ primitive directions $(s, 1)$ and $(1, 0)$; the forms λ'_s with $s \in p\mathbb{Z}/d, s \neq 0$, give the remaining $p^{k-1} - 1$ directions $(1, t)$ with $p \mid t \neq 0$. Hence the family $\{I\} \cup \{U_s : s \in \mathbb{Z}/d\} \cup \{V_s : s \in p\mathbb{Z}/d \setminus 0\}$ realises each of the $p^{k-1}(p+1)$ primitive directions exactly once. (This closes the first gap of the review: the family $F' Q_s$ alone, as in [3W], realises only $p^k + 1$ directions when $k \geq 2$; the V_s are needed, and they are what the verification script always used.) All phases lie in μ_d , or in μ_{2d} when $p = 2$.

The phases for $p = 2$. Let $d = 2^k$, $K \ni \omega_{2d}$ and $\mathcal{O}$ admissible. Then $2 \in \mathfrak{m}$, and 2 is a unit multiple of $(1 - \omega_{2d})^{2^k}$ in $\mathbb{Z}[\omega_{2d}]$ (since $1 - \omega_{2d}$ generates the totally ramified prime above 2); as $\mathfrak{m}$ is prime, $1 - \omega_{2d} \in \mathfrak{m}$. So every phase in μ_{2d} reduces to 1, exactly as the d -th roots of unity do. If the coefficient field does not contain ω_{2d} one adjoins it and extends the valuation ring ([St, Tag 00IA]); ranks are unchanged. This closes the second gap of the review.

For each U in the family put $C_U = UCU^{-1} = \sum_g c_g \varepsilon_g W_{M_{Ug}}$: it is an $\mathcal{O}$ -combination of Weyl matrices, hence lies in $M_d(\mathcal{O})$ although U is not unimodular, and $\text{rank}_K C_U = \text{rank}_K C$.

4.4 Theorem B'

Let λ be a primitive direction with $m_\lambda \geq 1$ and U the member of the family whose label map has first coordinate λ . Modulo $\mathfrak{m}$ every Z^b and every phase becomes I , while X becomes the cyclic shift $\bar{X}$ of order p^k over $\mathbb{k}$. Therefore

$$\bar{C}_U = \sum_{t \in \mathbb{Z}/d} \gamma_t^\lambda \bar{X}^t = \bar{g}_\lambda(\bar{X}), \quad \bar{g}_\lambda(x) = \sum_t \gamma_t^\lambda x^t \in \mathbb{k}[x],$$

with exponents taken in $\{0, \dots, d-1\}$ (legitimate since $\bar{X}^d = I$). The polynomial $\bar{g}_\lambda$ is nonzero, has m_λ terms and degree $< p^k$; as in Section 4.2 the nullity of $\bar{g}_\lambda(\bar{X})$ on $\mathbb{k}^d \cong \mathbb{k}[x]/(x-1)^{p^k}$ equals $\text{ord}_1 \bar{g}_\lambda$, which is at most $\Omega_k(m_\lambda)$ by Theorem A. Hence

$$\dim \ker_{\mathbb{C}} C = d - \text{rank}_K C_U \leq d - \text{rank}_{\mathbb{k}} \bar{C}_U = \text{ord}_1 \bar{g}_\lambda \leq \Omega_k(m_\lambda) \leq \Omega_k(|S|),$$

the last step because m_λ is at most the number of fibres meeting S and Ω_k is nondecreasing. $\square$

4.5 Lemma R

Suppose $f \neq 0$ has zero sum on every affine line and $|\text{supp } f| \leq 2p - 1$. A line meeting the support meets it in at least two points (a single nonzero value cannot sum to zero). Call one direction vertical and let $E \subseteq \mathbb{F}_p$ be the set of x whose column $\{x\} \times \mathbb{F}_p$ meets the support; $|E| = p$ would give at least $2p$ points, so $|E| \leq p - 1$, and $E \neq \emptyset$.

Let $R = \mathbb{k}[\mathbb{Z}/p] = \mathbb{k}[t]/(t^p - 1) = \mathbb{k}[u]/(u^p)$ with $u = t - 1$, acting on functions on $\mathbb{F}_p$ by translation, $(tg)(y) = g(y - 1)$, and let $g_x = f(x, \cdot) \in R$. A zero column sum means $g_x \in uR$ (the augmentation ideal). For a slope $\mu \in \mathbb{F}_p$, the sums over the lines $y = \mu x + c$ are the values of $\sum_{x \in E} t^{-\mu x} g_x$ at c , so

$$\sum_{x \in E} t^{\mu x} g_x = 0 \quad \text{in } R \text{ for every } \mu \in \mathbb{F}_p.$$

Let $v = \min_{x \in E} \text{ord}_u(g_x)$; then $1 \leq v \leq p - 1$ and $g_x = u^v g'_x$ with $g'_x(0) \neq 0$ for at least one x . Now $t^{\mu x} = (1 + u)^{\mu x} = \sum_{i < p} \binom{\mu x}{i} u^i$, and for $i < p$ the coefficient $\binom{n}{i} \bmod p$ depends only on $n \bmod p$ (Lucas) and equals $P_i(n)$ for the polynomial $P_i(n) = n(n-1) \cdots (n-i+1)/i!$ of degree i over $\mathbb{F}_p$. The coefficient of u^{v+j} ($0 \leq j \leq p - v - 1$) in the identity is $\sum_{x \in E} \sum_{i \leq j} P_i(\mu x) g'_{x,j-i}$, a polynomial in μ of degree $\leq j < p$ that vanishes on all of $\mathbb{F}_p$ and is therefore identically zero; its coefficient of μ^j comes only from $i = j$ and gives $\sum_{x \in E} x^j g'_x(0) = 0$ for $0 \leq j \leq p - v - 1$. If $p - v \geq |E|$ this is an invertible Vandermonde system in the distinct $x \in E$, forcing all $g'_x(0) = 0$, a contradiction; hence $v \geq p - |E| + 1$.

By Frenkel's lemma (the $k = 1$ case of Theorem A: a nonzero element of $u^v R$, viewed as a polynomial of degree $< p$ in t with order $\geq v$ at $t = 1$, has at least $v + 1$ nonzero coefficients), every nonempty column carries at least $v + 1 \geq p - |E| + 2$ support points. In particular $|E| = 1$ is impossible, since a column has only p points (equivalently, $v \geq p$ contradicts $v \leq p - 1$). Thus $2 \leq |E| \leq p - 1$ (so $p \geq 3$; for $p = 2$ we have already reached a contradiction) and

$$|\text{supp } f| \geq |E|(p + 2 - |E|) \geq \min(2p, 3(p - 1)) = 2p,$$

the minimum of the concave function over $[2, p - 1]$ being at an endpoint. This contradicts $|\text{supp } f| \leq 2p - 1$. (The review pointed out that minimising over $[1, p - 1]$ would only give $p + 1$; excluding $|E| = 1$ first, as above, is the missing line.) Sharpness: for parallel lines L_1, L_2 , a line of their class sums to $p \cdot (\pm 1) = 0$ or to 0, and any other line meets each of L_1, L_2 once, giving $1 - 1 = 0$. $\square$

4.6 Lemma N

Let $\bar{c} \neq 0$ have zero sum on every primitive line and suppose $|\text{supp } \bar{c}| \leq 2p - 1$. Fix g_0 in the support. Each of the $p^{k-1}(p + 1)$ primitive lines through g_0 contains another support point. A point $h \neq g_0$ with $h - g_0 = p^v h'$, h' primitive, $0 \leq v \leq k - 1$, lies on exactly p^v primitive lines through g_0 : the cyclic subgroups of order p^k containing $p^v h'$ are the $\langle h'' \rangle$ with h'' a lift to $\mathbb{P}^1(\mathbb{Z}/p^k)$ of the class of h' in $\mathbb{P}^1(\mathbb{Z}/p^{k-v})$, and there are $p^{k-1}(p + 1)/(p^{k-v-1}(p + 1)) = p^v$ of them. Let n_1 be the number of support points $h \neq g_0$ with $v = k - 1$, i.e. $h \in g_0 + T_1$, and n_2 the number with $v \leq k - 2$. Counting incidences,

$$n_1 p^{k-1} + n_2 p^{k-2} \geq p^{k-1}(p + 1), \quad n_1 + n_2 \leq 2p - 2,$$

whence $n_1(p - 1) \geq p^2 - p + 2$ and $n_1 \geq p + 1$ (for $p = 2$, $n_1 \geq 4$). So every T_1 -coset that meets the support contains at least $p + 2$ support points; two such cosets would give $2p + 4 > 2p - 1$ points, so the support lies in a single coset $g_0 + T_1$. A cyclic subgroup H of order p^k meets T_1 in its order- p subgroup $p^{k-1}H$, every order- p subgroup of $T_1 \cong \mathbb{F}_p^2$ arises this way, and a coset of H meets $g_0 + T_1$ either trivially or in a coset of $p^{k-1}H$, i.e. in an affine line of the plane $g_0 + T_1$; conversely every affine line of that plane is such an intersection. Since the support lies in the plane, the primitive-line sums are exactly the affine-line sums of the restriction of $\bar{c}$ to $g_0 + T_1 \cong \mathbb{F}_p^2$, and Lemma R gives $|\text{supp } \bar{c}| \geq 2p$, a contradiction. Sharpness: two parallel lines in $g_0 + T_1$ have zero sum on every affine line of the plane, hence on every primitive line. $\square$

4.7 Theorem B

Let $C = \sum_{g \in S} c_g W_g$ with $|S| = \ell \leq 2p - 1$ and all $c_g \neq 0$. Let $K = \mathbb{Q}(\omega_{2d})(c_g : g \in S)$ (for odd p , $\mathbb{Q}(\omega_{2d}) = \mathbb{Q}(\omega_d)$), a finitely generated field, and let $\mathcal{O}$ be a valuation ring of K extending $\mathbb{Z}[\omega_{2d}]_{(1-\omega_{2d})}$ (Chevalley); it is admissible, and for $p = 2$ it contains $1 - \omega_{2d}$ in its maximal ideal by construction. Scale by a coefficient of minimal valuation. The residue vector $\bar{c}$ is nonzero with support in S , of size at most $2p - 1$, so by Lemma N some primitive line carries a nonzero sum: there is a direction λ with $m_\lambda \geq 1$. Theorem B' gives $\dim \ker C \leq \Omega_k(m_\lambda) \leq \Omega_k(\ell)$, and the closed forms of Section 2 give the

explicit bounds. For the sharpness family, $C_\ell = f(Z)$ with $f(z) = \prod_{i=0}^{\ell-2} (z^{p^{k-1}} - \omega_p^i)$; by the q -binomial theorem ($\ell - 1 \leq p - 1$) f has exactly ℓ terms, and it vanishes at the $(\ell - 1)p^{k-1}$ elements of μ_d whose p^{k-1} -th power is one of $1, \omega_p, \dots, \omega_p^{\ell-2}$; so $s_W(C_\ell) = \ell$ and $\dim \ker C_\ell = (\ell - 1)p^{k-1} = \Omega_k(\ell)$. $\square$

4.8 Why $2p - 1$, and what the bound explains

At $\ell = 2p$ the combination $C = \sum_{g \in L_1} W_g - \sum_{g \in L_2} W_g$, for two parallel lines in a T_1 -coset, has integer coefficients whose residue vector lies in the obstruction code of Lemma N: every m_λ vanishes and Theorem B' is silent. The first-order method therefore stops exactly at $2p - 1$; whether the *conclusion* survives is Conjecture C.

For $\ell \leq 3$ the corollary of Section 3 says that the only orders of vanishing available to the reduced circulant are $0, p^j, 2p^j$ (resp. $0, 2^j$): the digit weight explains the shape of the corank set $\mathcal{N}_p$ of [0WE] Theorem 3, though Theorem B' by itself only bounds the nullity by such an order and does not force the nullity to be one. For $p < \ell \leq 2p - 1$ the bound $(p - 1)p^{k-1}$ is strictly stronger than the field-free bound $\text{rank } C \geq \lfloor d/\ell \rfloor$ ([3W] Section 5.2) as soon as $\ell > p$. Whether $\Omega_k(\ell)$ is attained at *exact* sparsity ℓ in this range is open: the family C_ℓ stops at $\ell = p$, and collinear supports do not attain it (for $d = 27$, $\ell = 4$ the bound is 18 while the collinear maximum at exact sparsity 4 is 12, Section 5.3). Ω_k is the monotone envelope, and Theorem Z shows that it is attained at exact sparsity $w_p(\Omega_k(\ell))$.

5. Composite dimension

5.1 Binomials

Proof of Theorem T2. Up to a nonzero factor and a phase, $C = c_1 W_g + c_2 W_h = c_1 W_g(I + cW)$ with $W = W_{h-g}$, $c \neq 0$. If $u = h - g$ has order $r \geq 2$ in G , then W^r is a scalar θI and $\text{Tr } W^j = 0$ for $r \nmid j$. The eigenvalues of W are the r roots of θ , with multiplicities m_i satisfying $\sum_i m_i \zeta_r^{ij} = 0$ for $0 < j < r$, so all $m_i = d/r$. The nullity of $I + cW$ is the multiplicity of $-1/c$, i.e. 0 or d/r , and r ranges over the orders of nonzero elements of $(\mathbb{Z}/d)^2$, all of which divide d ; the maximum $d/p_{\min}$ is attained by $I - Z^{d/p_{\min}}$. $\square$ (In polynomial form, $1 + cx^s$ vanishes on a fibre of $x \mapsto x^s$ on μ_d , of size $\gcd(s, d)$; this matches the two-label criterion of [0WE].)

5.2 Trinomials

Proof of Theorem T3, assuming [0WE] Lemma 1 and Theorem 2. After right multiplication by a Weyl adjoint, $D = cI + aU + bV$ with $abc \neq 0$ and $U = W_u, V = W_v$ of distinct nonzero labels. With $UV = \zeta VU$, $q = \text{ord } \zeta$, $H = \langle u, v \rangle$, $h = |\langle qu, qv \rangle|$, [0WE] Lemma 1 gives $|H| = q^2 h$ and a common sector multiplicity $m = d/(qh) \in \mathbb{Z}_{>0}$, and [0WE] Theorem 2 (4b) gives $\dim \ker D \leq m \min(2, h)$. If $h \geq 2$: $\dim \ker D \leq 2d/(qh)$, where qh divides d ; if $q = 1$ then $qh = |H| \geq 3$, and if $q \geq 2$ then $qh \geq 4$; so $qh \geq r_3(d)$ and $\dim \ker D \leq 2d/r_3$. If $h = 1$: $|H| = q^2$ forces $q \geq 2$, and $\dim \ker D \leq m = d/q \leq d/2$ with $q \mid d$, so d is even. Attainment: (i) let u have order r_3 and choose a phase so that $T = \xi' W_u$ satisfies $T^{r_3} = I$; its eigenvalues are the r_3 -th roots of unity, each of multiplicity d/r_3 ; with $\xi = e^{2\pi i/r_3}$, $(T - I)(T - \xi I) = T^2 - (1 + \xi)T + \xi I$ has three nonzero Weyl coefficients (labels $0, u, 2u$ distinct since $r_3 \geq 3$, and $1 + \xi \neq 0$) and nullity $2d/r_3$ — this is the construction of [0WE] Theorem 3 with p^j replaced by d/r_3 . (ii) If $d \equiv 2 \pmod{4}$, $U = Z^{d/2}$ and $V = X^{d/2}$ are Hermitian involutions with $UV = \omega^{d^2/4} VU = (-1)^{d/2} VU = -VU$, so $(U + V)^2 = 2I$ and $\text{Tr}(U + V) = 0$: $U + V$ has eigenvalues $\pm\sqrt{2}$ with multiplicity $d/2$ each, and $\sqrt{2}I + U + V$ has $s_W = 3$ and nullity $d/2$ (here $q = 2, h = 1$,

$m = d/2$). If $4 \mid d$ then $r_3 \leq 4$ and $2d/r_3 \geq d/2$ already. The “ ≤ 3 ” statement adds Theorem T2 (note $\frac{d}{2}[d \text{ even}] \leq d/p_{\min}$). $\square$

For $d = 10$ and 14 the anticommuting branch wins: $v_3 = 5$, resp. 7 , whereas the collinear maximum at exact sparsity three is $Z(10, 3) = Z(14, 3) = 4$ (exact, Section 5.3). Collinear supports are therefore not extremal at fixed sparsity, but they are extremal in the monotone sense: $\max_{s_W \leq 3} = \max(d/p_{\min}, 2d/r_3)$ is attained on the Z -axis. The [0WE] examples in $d = 12$ and 36 illustrate the sector mechanism and are not maxima: Theorem T3 gives $v_3(12) = 8$ and $v_3(36) = 24$. Numerically (sector formula and SVD, $d \in \{6, 8, 9, 10, 12, 14, 15\}$) the exact-sparsity-three maxima are $4, 4, 6, 5, 8, 7, 10$, all equal to the formula, with the $d = 10, 14$ witnesses anticommuting.

5.3 What fails for composite d , what survives, and the exact tables

If d has two distinct prime factors then $\Phi_d(1) = 1$, $1 - \omega_d$ is a unit and no prime of $\mathbb{Z}[\omega_d]$ has $\omega_d \equiv 1$: the valuation argument has no starting point ([3W] Section 5.4). No additive law in the largest prime factor $p_{\max}$, or largest prime-power factor P , of d can hold: at $\ell = 2$ any bound of the form $(\ell - 1)d/p_{\max}$ or $(\ell - 1)d/P$ is $< d/p_{\min}$, the true maximum of Theorem T2, whenever d is not a prime power ($d = 10$: bound 2, truth 5; $d = 12$: bound 4 or 3, truth 6; at $\ell = 3$, $d = 12$: truth $8 = 2d/3$). The extremal parameters are the smallest divisors $p_{\min}$ and r_3 , in line with Lam–Leung [LL00]: a vanishing sum of m d -th roots of unity exists only if m is a nonnegative combination of the prime factors of d . What survives for all d is the field-free bound $\text{rank } C \cdot s_W(C) \geq d$ and, for collinear supports, Meshulam’s inequality [Me06] on $\mathbb{Z}/d$; the latter is not attained in general: on $\mathbb{Z}/10$ it allows 6 zeros at $m = 3$ and 7 at $m = 4$, while the exact values are 4 and 6.

Exact tables. $Z(d, m)$ was computed by enumerating, for each support S with $|S| = m$ up to the affine group of $\mathbb{Z}/d$ (units and translations), the flats cut out by at most $m - 1$ character rows over $\mathbb{Q}(\omega_d)$, with arithmetic in $\mathbb{Q}(\omega_d)$ realised through the regular representation and kernels over $\mathbb{Q}$; a flat carries a full-support vector if and only if no coordinate vanishes identically on it, and its closure is then a zero set. (Correctness: a full-support f with zero set T spans a subspace of dimension $\leq m - 1$ of row conditions, so some $R \subseteq T$ with $|R| \leq m - 1$ cuts the same flat, and every full-support vector of that flat vanishes on the closure.) The reviewer recomputed the values by a different exact method (rank criterion over pairs (S, T) , refuting $v + 1$ zeros and attaining v , with python-flint). Only the entries certified in this way are called exact; the remaining entries come from the earlier finite-field model ($\mathbb{F}_q$ with two independent primes $q \equiv 1 \pmod d$, which can only under-estimate ranks and is therefore not a certificate in either direction at fixed m , although two agreeing primes make an error very unlikely). The working report had labelled all of them exact; that label is withdrawn here.

d	exact over $\mathbb{Q}(\omega_d)$ ($m = 1, 2, \dots$)	$\mathbb{F}_q$ only (not certified)
6	0, 3, 4, 4, 4, 5 ($m \leq 6$)	—
10	0, 5, 4, 6, 8, 7, 6, 8, 8, 9 ($m \leq 10$)	—
12	0, 6, 8, 9, 8, 10, 8 ($m \leq 7$)	$m = 8$: 10
14	0, 7, 4, 8, 8 ($m \leq 5$)	—
15	0, 5, 10, 9, 12 ($m \leq 5$)	$m = 6, 7$: 11, 9
18	0, 9, 12, 12 ($m \leq 4$)	—

For $d = 15$ the values $m \leq 3$ follow from Theorems T2, T3 (collinear attainment), $m = 4$ is the reviewer’s exact spot check, and $m = 5$ is an exact run of the author’s flats script made for this manuscript

(34 support orbits, 4455 s; repro/out/zerosQ_cyclic_15_m5.out). Entries for $d = 20, 21, 30$ mentioned in the working report have no log and are dropped. Constructions matching the exact entries: $m = 2$: a fibre of $x^{d/p_{\min}}$; $m = 3$: $2d/r_3$ ($d = 6, 12, 15, 18$) or, for $d = 10, 14$, the value 4; $m = 4$: $3d/r_4$ with r_4 the smallest divisor ≥ 4 ($d = 12$: 9; $d = 18$: 12), or the product $(1 + ax^{d/r})(1 + bx^{d/s})$ with $\gcd(r, s) = 1$, which has $d(r + s - 1)/(rs)$ zeros ($d = 6$: 4; $d = 10$: 6; $d = 14$: 8); $m = 5, d = 10$: $(m - 1)d/5 = 8$. No closed formula for general m is claimed.

For prime powers the running maxima are Theorem Z; the exact-sparsity values are not monotone. For $d = 16$ the values $Z(16, 3) = 8$ and $Z(16, 4) = 12$ are certified exactly (reviewer), and $Z(16, m) = \Omega_4(m)$ for $m \leq 2$ by Theorem Z; the $\mathbb{F}_q$ model gives 8, 12, 12 for $m = 5, 6, 7$. For $d = 27$, $Z(27, m) = \Omega_3(m)$ for $m \leq 3$ by Theorem Z, and $Z(27, 4) = 12$ is certified exactly by the reviewer's flats method (Section 9) — this is the value used in Section 4.8.

6. Conjecture C and the evidence

6.1 Where the conjecture is open

For $d = p^k$ and a sparsity ℓ , Conjecture C is implied by Theorem B when $\ell \leq 2p - 1$ and by the field-free bound when $\Omega_k(\ell) \geq d - \lfloor d/\ell \rfloor$. The remaining sparsities, computed by two independent scripts (author and reviewer, identical output), are:

d	Theorem B covers $\ell \leq$	open ranges of ℓ (rank that must be proved)
4, 8	3	none: Conjecture C is a theorem
9	5	none: Conjecture C is a theorem
16	3	[6, 7] (rank ≥ 4 ; field-free gives 3)
25	9	[10, 14] (≥ 4); [15, 19] (≥ 3)
27	5	[6, 8] (≥ 6); [14, 17] (≥ 3)
32	3	[5, 7] (≥ 8); [11, 15] (≥ 4)
49	13	[14, 20] (≥ 6); [21, 27] (≥ 5); [28, 34] (≥ 4); [35, 41] (≥ 3)

The first open case overall is $d = 16, \ell = 6$: is every $C \in M_{16}(\mathbb{C})$ with six or seven nonzero Weyl coefficients of rank at least 4?

6.2 Torsion cosets

Let $2j \leq k$ and let the support lie in a coset of T_j . After a translation the support lies in $T_j = p^{k-j}G$, where $\langle g, h \rangle \in p^{2(k-j)}\mathbb{Z} = 0$ in $\mathbb{Z}/d$, so $g \mapsto W_g$ is an honest representation of $T_j \cong (\mathbb{Z}/p^j)^2$; by trace orthogonality each character χ occurs with multiplicity p^{k-2j} , and $C = \sum_{\chi} \hat{c}(\chi)P_{\chi}$ with $\hat{c}(\chi) = \sum_g c_g \chi(g)$. Hence $\dim \ker C = p^{k-2j} \cdot \#\{\chi : \hat{c}(\chi) = 0\}$, a commutative uncertainty problem on $(\mathbb{Z}/p^j)^2$.

p -torsion ($j = 1, k \geq 2$). Meshulam's inequality on $\mathbb{F}_p^2$ with the consecutive divisors $1, p, p^2$ gives, for $\ell \leq p$, at most $p(\ell - 1)$ zero characters, and for $p \leq \ell \leq p^2$ at most $p^2 - p - 1 + \lfloor \ell/p \rfloor$; multiplied by p^{k-2} these are exactly $(\ell - 1)p^{k-1}$ and $(p - 1)p^{k-1} + (\lfloor \ell/p \rfloor - 1)p^{k-2}$, i.e. $\Omega_k(\ell)$ (with $\Omega_k(p^2) = p^k - p^{k-2}$

at $\ell = p^2$). So Conjecture C holds for every support in a coset of the p -torsion, for every $k \geq 2$. (Re-derived by the reviewer; the comparison script reports no case with $j = 1$ in which the Meshulam bound exceeds $\Omega_k(\ell)$, over all $p \leq 7$, $p^k \leq 2401$ and $\ell \leq p^2$, i.e. 316 cases.)

4-torsion in $d = 16$ and 32 ($p = 2, j = 2$). Here Meshulam's bound on $(\mathbb{Z}/4)^2$ is slack (it allows $10 > 8$ zeros at $\ell = 3$ and $13 > 12$ at $\ell = 6, 7$), so exact values are needed. The exact maxima of zero characters on $(\mathbb{Z}/4)^2$ for full-support functions with exactly m points are 0, 8, 8, 12, 8, 12, 10 for $m = 1, \dots, 7$ (author, exhaustive over the orbits of the affine group of order 1536; reviewer, independent method, $m \leq 6$). Fourier duality closes the rest: $|\text{supp } f| \in \{2, 3\}$ forces $|\text{supp } \hat{f}| \geq 8$, hence (applying this to $\hat{f}$) $|\text{supp } \hat{f}| \in \{2, 3\}$ forces $|\text{supp } f| \geq 8$, and $|\text{supp } \hat{f}| = 1$ forces $|\text{supp } f| = 16$. Therefore $|\text{supp } f| \leq 7$ gives $|\text{supp } \hat{f}| \geq 4$, at most 12 zeros, and $8 \leq |\text{supp } f| \leq 15$ gives at most 14 zeros. Since $\Omega_4(m) = 12$ for $4 \leq m \leq 7$ and 14 for $8 \leq m \leq 15$, Conjecture C holds for every support in a coset of T_2 in $d = 16$ (multiplicity 1), and, with multiplicity $2^{5-4} = 2$, in $d = 32$: the bounds 16, 24, 28, 30 for $m \in \{2, 3\}, [4, 7], [8, 15], \{16\}$ equal Ω_5 there.

6.3 Numerical searches and a false alarm

Alternating projections between the span of the support Weyls and the variety of rank- $\leq d - \kappa$ matrices were run in the open ranges: $d = 16, \ell \in \{5, 6, 7\}$, target nullity 13 (300 random and 100 “blocked” supports containing a T_1 -coset, 6 restarts, 300 iterations each); $d = 27, \ell \in \{6, 7, 8\}$, target 22; $d = 25, \ell \in \{10, 12, 14\}$ (target 22) and 16 (target 23). No full-support witness was found (best relative residuals $\geq 0.46, 0.35, 0.18$). Control searches against the proved statements ($d = 9, \ell = 4, 5$; $d = 8, \ell = 3$; $d = 16, \ell = 3$; $d = 25, \ell = 8, 9$; $d = 27, \ell = 4, 5$) likewise found nothing, as Theorem B requires. These are finite searches, not evidence of the same kind as Sections 6.1–6.2.

Three instances in $d = 16$ had been flagged numerically as having nullity 1 where Theorem B' predicts 0 (a single coefficient survives modulo 2). Exact computation of the rank of the 128×128 integer matrix of the regular representation gives 128 in all three cases, so $\det C \neq 0$; the matrices are merely ill-conditioned ($\sigma_{\min}/\sigma_{\max}$ between 10^{-16} and 10^{-12} , the sector determinant $c^{16} - (a^{16}s + b^{16}t)$ nearly cancelling). The reviewer confirmed the first instance by hand: its relative labels have commutator of order 16, one sector, and determinant $1 - 8^{16}(s + t)$ with s, t roots of unity, which cannot vanish. No violation of Theorem B' exists.

7. Verification

Every statement below was checked by the author's scripts and, independently, by the reviewer's (Section 9 lists commands and runtimes).

1. *Theorem A*. Exhaustive over exponent sets (up to translation, resp. up to the affine group of $\mathbb{Z}/p^k$), with the maximal order over full-support polynomials decided by a rank criterion on the Lucas/Hasse matrix over $\mathbb{F}_p$ (the statement over the algebraic closure): $d = 16$ all m ; $d = 25, 27$ for $m \leq 7$ (author); $d = 4, 8, 9, 16$ all m and $d = 25, 27$ for $m \leq 7$ (reviewer, both the exact-order form $w_p(\text{ord}) \leq \# \text{terms}$ and the monotone form). The non-monotone exact maxima are reproduced ($d = 16$: exactly 3 terms give at most order 4, exactly 2 give 8).
2. *Theorem Z*. Exact certification in $\mathbb{Z}[\omega_d]$ (arithmetic modulo Φ_d) that f_r has exactly $w_p(r)$ terms and at least r zeros in μ_d , for every $r < d$ and $d \in \{4, 8, 9, 16, 25, 27, 32, 49, 81\}$ (author; 1–95 s per d); exactly r zeros for $d \in \{4, 8, 9, 16, 25, 27\}$ (reviewer, own arithmetic). Also the sharpness family C_ℓ .

3. *Clifford identities*. Author: all conjugation identities for U_s and V_s checked numerically (complex doubles, discrete target, tolerance 10^{-9}) for $d = 16, 25, 27$ (4352, 15625, 19683 identities): no failure. Reviewer: exact verification in $\mathbb{Z}[\omega_{2d}]$ ($p = 2$) or $\mathbb{Z}[\omega_d]$ for $d = 4, 8, 9$ (96, 768, 972 identities), including that the phases lie in μ_{2d} resp. μ_d and that the family realises exactly 6, 12, 12 directions.
4. *Lemmas R and N*. Reviewer: minimum weight of the Radon code on $\mathbb{F}_p^2$ for $p = 2, 3, 5$ and of the obstruction code $N(p, k)$ for $d = 4, 8, 9, 16$ equals $2p$ (meet-in-the-middle over column combinations, exact over $\mathbb{F}_p$, field-extension invariant). Author: $\dim N$ over $\mathbb{F}_p$ is 7, 34, 45, 160, 400, 489 for $d = 4, 8, 9, 16, 25, 27$, the two-line word lies in N , the full weight distribution for $d = 4$ (127 words, minimum 4), and an independent minimum-distance check for $d = 8, 16, 9$.
5. *Theorem B' on instances*. 300 ($d = 16$) and 200 ($d = 25, 27$) integer-coefficient instances, comparing the numerical nullity with $\text{ord}_1 \bar{g}_\lambda$ in every primitive direction; the three flags were resolved exactly (item 6.3).
6. *Theorems T2, T3*. Sector formula plus SVD for $d \in \{6, 8, 9, 10, 12, 14, 15\}$ (author); exhaustive binomial maxima for $2 \leq d \leq 15$, 60 random trinomials against the sector formula, and the attainment examples (reviewer).
7. *Tables and gap ranges*. Section 5.3 and 6.1: author and reviewer values agree wherever both exist; the gap table and the closed forms of Ω_k were recomputed independently for all listed d .

8. Open questions

1. **Conjecture C at $d = 16, \ell \in \{6, 7\}$** . A proof would need either a second-order (Newton-polygon) refinement of Theorem B' for residue vectors in the obstruction code, or an exact exhaustive computation over the $\text{AGL}(2, \mathbb{Z}/16)$ -orbits of 6- and 7-point supports, which is within reach of the flats method.
2. **Exact sparsity**. Is $\Omega_k(\ell)$ attained at exact sparsity ℓ for $p < \ell \leq 2p - 1$? The collinear answer is no ($d = 27, \ell = 4$: $12 < 18$); the non-collinear answer is unknown, and by [OWE] Theorem 3 the coranks at $\ell = 3$ are constrained to $p^j, 2p^j$.
3. **Composite $d, \ell \geq 4$** . Conjecture U and a closed form for $Z(d, m)$; the product constructions of Section 5.3 and the values $3d/r_4$ suggest that the smallest divisors continue to govern, but $Z(10, 5) = 8 = 4d/5$ and $Z(15, 5) = 12 = 4d/5$ are already of the form $(m - 1)d/r$ with $r = 5$ the *largest* divisor below $m + 1$.
4. **The obstruction code**. Determine the minimum-weight words and the weight distribution of $N(p, k)$ beyond the two-line words; a characterisation of low-weight words would tell exactly where Theorem B' can and cannot be applied.
5. **Other groups**. The proof uses only that $1 - \omega$ is a prime with residue characteristic p and that the Clifford family covers all primitive directions; the same argument should apply to Weyl systems over $(\mathbb{Z}/p^k)^{2n}$ and, with Meshulam's inequality replacing Theorem A, to groups of order pq in the spirit of [Lo25].

9. Reproducibility

The directory `repro/` next to this manuscript contains: `author/` (the thirteen scripts of the working report, unchanged, with their logs in `author/logs/`); `reviewer/` (the reviewer's nine scripts and two logs, unchanged); `run_fast.py`, which reruns the fast subset and writes `out/<name>.out` and `out/runtimes.txt`; and `README.md` with provenance and the list of slow runs that were not repeated (the exact tables for $d = 10, 12, 14, 18$ and the plane $(\mathbb{Z}/4)^2$ beyond $m = 4$, the $d = 25, 27$ obstruction

codes, the numerical searches). Python 3.12, numpy 2.5.1, sympy 1.14, python-flint 0.9.0 (reviewer's exact scripts only). The rerun of 13 September 2026 is summarised in `README.md`; every fast check reproduced the logged result.

AI-assistance statement

This manuscript, the mathematics and the code were produced by Claude Fable 5.1 agents (Anthropic) working under the author's direction: two author agents (working report of 12–13 September 2026, the second resuming the first after a usage limit; no fallback to another model occurred), an independent adversarial review agent (13 September 2026), whose three expository gaps (G1: the V_s family; G2: the $p = 2$ phases; G3: the case $|E| = 1$ in Lemma R) and labelling correction (E1: exact versus $\mathbb{F}_q$ tables) are incorporated in Sections 4.3, 4.5 and 5.3 with attribution, and a writing agent (this text, `run_fast.py`, the rerun). Theorem A is a known result [MCJ73]; Theorem T3 depends on [OWE] Theorem 2 as published and was not re-proved. No human peer review has taken place. Scientific responsibility rests with the author.

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